Editor	ZsRE				CounterFact			
	Score	Efficacy	Generalization	Locality	Score	Efficacy	Generalization	Locality
— (original GPT-J)	26.4	26.4 (0.6)	25.8 (0.5)	27.0 (0.5)	22.4	15.2 (0.7)	17.7 (0.6)	83.5 (0.5)
FT-W (21st layer w/ weight decay)	42.1	69.6 (0.6)	64.8 (0.6)	24.1 (0.6)	67.6	99.4 (0.1)	77.0 (0.7)	46.9 (0.6)
MEND (Mitchell et al., 2022)	20.0	19.4 (0.5)	18.6 (0.5)	22.4 (0.5)	23.1	15.7 (0.7)	18.5 (0.7)	**83.0** (0.5)
ROME (Meng et al., 2022)	2.6	21.0 (0.7)	19.6 (0.7)	0.9 (0.1)	50.3	50.2 (1.0)	50.4 (0.8)	50.2 (0.6)
MEMIT (Meng et al., 2023)	50.8	96.7 (0.3)	89.7 (0.5)	26.6 (0.5)	85.8	98.9 (0.2)	88.6 (0.5)	73.7 (0.5)
PMET (Li et al., 2024b)	51.0	96.9 (0.3)	90.6 (0.2)	26.7 (0.2)	86.2	99.5 (0.1)	92.8 (0.4)	71.4 (0.5)
FT	44.8	**99.9** (0.0)	**98.9** (0.2)	21.4 (0.5)	52.8	79.6 (0.8)	58.5 (0.8)	36.8 (0.7)
FT (21st layer)	42.9	**99.9** (0.0)	87.4 (0.5)	20.5 (0.5)	60.5	99.9 (0.04)	63.3 (0.8)	42.0 (0.6)
FT + Mask	58.3	97.6 (0.3)	91.7 (0.5)	32.9 (0.6)	54.3	97.1 (0.3)	62.1 (0.8)	34.7 (0.6)
FT + Mask + Para	56.1	**99.9** (0.0)	98.7 (0.2)	29.9 (0.5)	63.7	**100.0** (0.0)	92.5 (0.4)	38.0 (0.6)
FT + Mask + Para + Rand	**62.0**	**99.9** (0.0)	97.0 (0.3)	**35.6** (0.6)	**86.5**	98.8 (0.2)	**93.6** (0.4)	72.0 (0.6)
FT + Mask + Para + Rand + DPO	—	—	—	—	85.5	98.8 (0.2)	93.4 (0.4)	70.1 (0.6)

Table 2: Mass-editing results on ZsRE and CounterFact (10,000 edits each) with GPT-J. The results of FT-W, MEND, ROME, and MEMIT are from Meng et al. (2023). FT denotes naive fine-tuning on the requested edits; FT (21st layer) denotes FT only on the 21st layer. "+ Mask" means we mask the prompt. "+ Para" means paraphrase augmentation (which is involved in all the baselines). "+ Rand" means we augment the data with *random* facts from the training split (not overlapping with any evaluation facts). "+ DPO" means we additionally optimize the DPO loss term using the changed target as the preferred response over the pre-edit target (only provided in CounterFact). Except for FT (21st layer), all our results use LoRA for computational efficiency.

Editor	ZsRE				CounterFact			
	Score	Efficacy	Generalization	Locality	Score	Efficacy	Generalization	Locality
— (original GPT-2 XL)	22.5	22.2 (0.5)	21.3 (0.5)	24.2 (0.5)	30.5	22.2 (0.9)	24.7 (0.8)	78.1 (0.6)
FT	45.9	99.6 (0.1)	82.1 (0.1)	23.2 (0.5)	65.1	**100.0** (0.0)	87.9 (0.6)	40.4 (0.7)
FT-L	40.1	92.3 (0.4)	47.2 (0.7)	23.4 (0.5)	66.9	99.1 (0.2)	48.7 (1.0)	70.3 (0.7)
KN	-	-	-	-	35.6	28.7 (1.0)	28.0 (0.9)	72.9 (0.7)
KE	41.8	65.5 (0.6)	61.4 (0.6)	24.9 (0.5)	52.2	84.3 (0.8)	75.4 (0.8)	30.9 (0.7)
MEND	42.9	75.9 (0.5)	65.3 (0.6)	24.1 (0.5)	57.9	99.1 (0.2)	65.4 (0.9)	37.9 (0.7)
ROME	47.9	99.8 (0.0)	88.1 (0.5)	24.2 (0.5)	**89.2**	100.0 (0.1)	**96.4** (0.3)	**75.4** (0.7)
FT + Mask + Para + Sim	**51.4**	**100.0** (0.0)	**99.9** (0.0)	**26.1** (0.8)	83.1	98.6 (0.3)	87.3 (0.7)	69.0 (0.7)

Table 3: Single-editing results on ZsRE (10,000 edits) and CounterFact (7,500 edits) with GPT-2 XL. "+ Sim" means we include similar facts (measured by Sentence-BERT) instead of random facts for locality supervision.

We follow the setting in ROME and use GPT-2 XL (1.5B). The training takes around 1.4 seconds per edit on average. We give the results in Table 3. We find that it is helpful to include similar facts instead of random facts for locality supervision, presumably because fine-tuning is more sensitive with small data size thus requiring more care in the selection of demonstrations. We use Sentence-BERT embeddings (Reimers and Gurevych, 2019) and take 15 nearest facts as $\mathcal{R}$. On ZsRE, our method outperforms all existing single-edit methods. On CounterFact, our method lags behind ROME which is specifically optimized for single-editing, but still achieves the second-best edit score.

5.3 Generative Metrics

We report results on generative metrics in Table 4 (see Appendix B for single-editing results). Following prior work (Meng et al., 2022, 2023), we report fluency (entropy of n-gram distributions) and consistency (similarity score with a reference text). Our fine-tuning methods take a hit on these metrics while improving the edit score, showing that more work is needed to go beyond classification. However, recent work shows that *none* of the compared editors preserves downstream performance (Gu et al., 2024), thus achieving this goal meaningfully is still an open problem.

We additionally show that we can easily incorporate considerations for generative performance into fine-tuning. Specifically, we optimize $(1-\gamma)L_1(\theta) + \gamma L_2(\theta)$ where $L_1(\theta)$ is the loss in (2) and $L_2(\theta) = -\sum_{w \in \mathcal{W}} \log p_\theta(w)$ is a language modeling loss on random Wikipedia text. We choose Wikipedia articles so that they do not overlap with those used for consistency evaluation. We set $\gamma = 0.1$. With this change ("+ Wikipedia

Editor	Score	Fluency	Consistency
— (original GPT-J)	22.4	622.4 (0.3)	29.4 (0.2)
FT-W	67.6	293.9 (2.4)	15.9 (0.3)
MEND	20.0	618.4 (0.3)	31.1 (0.2)
ROME	50.3	589.6 (0.5)	3.3 (0.0)
MEMIT	85.8	619.9 (0.3)	40.1 (0.2)
PMET	86.2	620.0 (0.3)	40.6 (0.2)
FT	52.8	626.1 (0.4)	31.0 (0.2)
FT + Mask	54.3	563.6 (0.5)	6.1 (0.1)
FT + Mask + Para	63.7	550.7 (0.6)	4.7 (0.1)
FT + Mask + Para + Rand	86.5	352.0 (1.5)	5.2 (0.2)
+ Wikipedia Loss	84.8	609.2 (0.6)	29.2 (0.2)

Table 4: Fluency and consistency with mass-editing on COUNTERFACT. "+ Wikipedia Loss" means we additionally optimize a language modeling loss on Wikipedia text (Section 5.3).

Editor	Score	Efficacy	Generalization	Locality
— (GPT-2 XL)	29.9	21.8 (0.8)	24.1 (0.7)	78.3 (0.5)
ROME	50.4	50.3 (0.9)	49.4 (0.8)	51.6 (0.6)
MEMIT	71.5	79.9 (0.7)	66.2 (0.8)	69.8 (0.5)
FT + M + P + R	85.4	98.8 (0.2)	86.5 (0.5)	74.3 (0.5)
w/o LoRA	86.9	98.8 (0.2)	91.2 (0.4)	74.4 (0.5)

Table 5: Mass-editing results on COUNTERFACT with and without LoRA.

Loss"), we obtain a significant improvement in both fluency and consistency at the cost of a modest drop in the edit score.

5.4 Analysis

What is the effect of PEFT? We repeat mass-editing experiments with a smaller model (GPT-2 XL, 1.5B) to compare the performance of LoRA vs full fine-tuning in Table 5. We see that we achieve better generalization without LoRA, thus our improvement comes from conditional likelihood optimization and data augmentation, not the use of adapters (i.e., unlike specialized model editors that are based on adapters).

Does layer selection help? Layer selection is an important component in many existing model editing works. To see if it also benefits us, we fine-tune (without LoRA) only the layers 3–5 in GPT-J which are a subset of the layers chosen in MEMIT. Table 6 shows that we obtain a significant improvement compared to updating all layers (Table 2). Thus layer selection helps on top of our already strong fine-tuning approach.

Subject knowledge vs data augmentation. A common but rather strong assumption in many state-of-the-art editors is that the subject phrase in a fact is known. This knowledge critically allows the model to completely avoid changing at test time if the subject is not recognized, giving a significant advantage in locality. For instance, with perfect subject knowledge, locality is optimal. We apply our neighborhood fact augmentation to SWEA⊕OS (Li et al., 2024a) which holds a state-of-the-art result on COUNTERFACT using subject knowledge (see Appendix C for details). Table 7 shows that the augmentation does not help further. Hence subject knowledge obviates the need for data augmentation aimed at improving locality.

Editor	Score	Efficacy	Generalization	Locality
FT + M	63.0	98.1 (0.2)	61.2 (0.8)	47.4 (0.6)
FT + M + P	77.8	100 (0.0)	98.6 (0.2)	54.2 (0.6)
FT + M + P + R	91.1	98.8 (0.2)	96.8 (0.3)	80.2 (0.5)

Table 6: Mass-editing results on COUNTERFACT with GPT-J but only fine-tuning layers 3–5.

Editor	Score	Efficacy	Generalization	Locality
— (GPT-J)	22.4	15.2 (0.7)	17.7 (0.6)	83.5 (0.5)
SWEA⊕OS	91.2	99.5 (0.1)	98.1 (0.2)	79.0 (0.5)
SWEA⊕OS + $\mathcal{R}_\mathcal{E}$	91.2	99.5 (0.1)	97.7 (0.2)	79.4 (0.5)

Table 7: Mass-editing results (reproduced) on COUNTERFACT by adding neighborhood augmentation to SWEA⊕OS.

6 Conclusions

We have demonstrated that standard fine-tuning is sufficient to obtain strong edit performance with a slight modification: optimizing the conditional likelihood and augmenting the data with additional facts to promote locality. Our results challenge the assumption that standard fine-tuning is ineffective as a model editor, suggesting that model editing could be achieved as part of standard training rather than through a specialized model editor.

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Limitations

Our scope is limited to the classification aspect of model editing rather than the generative aspect to demonstrate the viability of a fine-tuning editor. Thus our approach is currently unable to preserve the LLM's general capabilities, as is the case with existing mass editors.

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A Mass-Editing on WikiRecent

To further validate our mass-editing results, we perform additional experiments on a simplified version of the WikiRecent dataset (Cohen et al., 2023). The original dataset consists of 570 and 1,266 facts from Wikidata for training and testing, where each fact has been recently modified (after July 2022). An example prompt is π ="The name of the position held by Nicolaus Bergius is" with the target object o ="bishop". Since the dataset does not provide paraphrase prompts for evaluating generalization, we create paraphrases of π by few-shot prompting GPT (`gpt-3.5-turbo-0125`). The prompt for GPT is given in Figure 2. Since the task is relatively straightforward, we find the generated paraphrases preserve the original meaning (e.g., "The position held by Nicolaus Bergius is"). For neighborhood prompts, we select 10 random